



Chondrocranium features of frog larvae of the genus *Physalaemus* and their relatives

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The genus *Physalaemus* comprises currently 42 species divided in seven species groups. Larvae of the genus is a typical “pond type”. Presently about 26 species assigned to the genus had its tadpole described, but few data is available about chondrocranium features. In addition, the diagnosis of the genus is fully based on adult traits. My intent is to make a character survey concerning chondrocrania of *Physalaemus* larvae in a way that allow it to be further included in the diagnosis of the genus. The neurocranium is usually longer than wide and depressed. The large amount of variation within *Physalaemus* is found in the cartilago suprarostal and in the processes of the palatoquadrate. The former has a V- or U-shaped corpus and is dorsally connected to the alae, which has an upper process. In the genera *Engystomops* and *Pleurodema*, the corpus of the suprarostal is fused ventrally to the alae. In *Eupemphix* larvae, this character is similar to the *Physalaemus* condition. The cornua trabeculae are usually narrow and wide, its length varies little. Tectum synoticum is a thin bar connecting the two otic capsulae. Otic capsulae are ovoid and bears a poorly developed anterolateral process. Fenestra frontoparietalis is always opened. The basihyal is always absent. The size of the processes of ceratohyalia varies slightly. The planumhypobranchiale shows a U-shaped posterior margin, although its extent can vary greatly. The processus anterolateralis is generally well developed. There are always three spiculae, associated to the ceratobranchials. The processus branchialis is only common in *Pleurodema*.

Keywords: *Physalaemus*, *Pleurodema*, *Eupemphix*, *Engystomops*, tadpole, chondrocranium, phylogeny, Amphibia, Anura, Neobatrachia